CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EE)

May 2014

EE 206 Digital Electronics & Microprocessor

Date: 09.05.2014, Friday | Time: 10:00 a.m. To 01:00 p.m. | Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Fill in the blanks [07]

i) The binary equivalent of decimal 11 will be ______________.

ii) Expression Y = (A+B)' represent ____________ gate.

iii) To construct a register to store 32-bit data, we will need a minimum of ___ flip-flops.

iv) A multiplexer has ____ inputs, ___ select lines and ___ output.

v) CMOS stands for _____________.

vi) Figure of merit = ____________ × power dissipation.

vii)_________ gate that will have HIGH or "1" at its output when any one of its inputs is HIGH.

Q - 2 (a) Simplify the following Boolean expression: [06]

i) Y = (A' + A) (B B') + (C' +1) (D' 0)

ii) Y = (AB' + A'B + AB + A'B')

iii) Y = (A+C) (A+D) (B+C) (B+D)

(b) Do as directed. [04]

i) Convert the binary 11010111 into its decimal equivalent.

ii) Convert decimal 3000 into its octal equivalent.

iii) Convert the binary number 1010 1111 1011 0010 into the equivalent hex number.

iv) Convert the hex number 4CA in to its equivalent octal form.

(c) Realize Ex-OR operation using only NAND gates. [04]

OR

Q - 2 (a) Explain a 4-bit bidirectional shift register. [06]

(b) Explain the features of Integrated –Injection Logic (IIL) and its operation. [04]

(c) With a neat circuit diagram explain the working of decimal to BCD encoder. [04]

Q - 3 (a) Write notes on positive level triggered S-R Flip-flop. [06]

(b) Reduce the following expression using K-Map [04]

$\sum m\,(0,1,2,3,10,11,14,15)$

(c) Construct a circuit using AND, OR and NOT gates for the following Boolean expression [04]

$Y = (A' + B' + C')\,(A + B + C)$

OR

Q - 3 (a) Implement the following function using an 8:1 multiplexer. [06]

$F(A,B,C,D) = \sum m(1,3,5,7,9,11,13)$

(b) Write a notes on ring counter. [04]

(c) Write the truth table of BCD to seven segment decoder and implement it using a 4 line to 16 line decoder. Assume the display to be common anode display. [04]

SECTION – II

Q - 4 (a) Answer the following questions: [04]

i) If the memory chip size is 256×1 bits, how many chips are required to make up 1 K (1024) bytes of memory ?

ii) What is the function of accumulator ?

iii) Why is the data bus bidirectional ?

iv) Write the machine code for the instruction MOV H,A if the opcode = 01_2, the register code for H = 100_2, and the register code for A = 111_2.

(b) Draw pin diagram of 8085 and give function of ALE and trap pin. [03]

Q - 5 (a) Explain the microprocessor initiated operation with the bus structure. [06]

(b) What is flag? Draw the format of 8085 flag register and explain in detail. [06]

(c) Draw the timing diagram of Memory Read operation. [02]

OR

Q - 5 (a) Draw architecture of 8085 microprocessor and explain register array [06]

(b) Write 8085 microprocessor based assembly language program to find the largest number in a data array. The number in a series are 99,74 and 96. The count is placed in the memory location 2500H. The numbers are placed in the memory location 2501 to 2503 H. The result is to be stored in the memory location 2450H. [06]

(c) Explain register addressing mode with examples. [02]

Q - 6 (a) Explain operation of following Instruction: [06]

I) LDA addr. II) CMA III) XCHG IV) ANA r
V) CALL addr. VI) ADD M

(b) Illustrate the steps and timing of data flow when the instruction code 0100 1111 (4FH – MOV C,A), stored in location 2005 H, is being fetched. [06]

(c) Write an assembly language program for addition of two 8-bit numbers. The 1st number 49H is in the memory location 3000H. The 2nd number 56H is in the memory location 3001H. Store the result in memory location 3002H. [02]

OR

Q - 6 (a) Do as directed: [06]

I) Why are the program counter and stack pointer 16-bit registers ?

II) The memory address of the last location of a 8K byte memory chip is FFFF H. Find the starting address.

III) Explain Electrically Erasable Programmable Read only Memory (EE-PROM).

(b) Writ a program to count from 0 to 9 with a one-second delay between each count. At the count of 9, the counter should reset itself to 0 and repeat the sequence continuously. Use register pair HL to set up the delay, and display each count at one of the output ports. Assume the clock frequency of the microcomputer is 1 MHZ. [06]

(c) Assume register B holds 93H and accumulator hold 15 H . Illustrate the results of the instruction ORA B and CMA. [02]